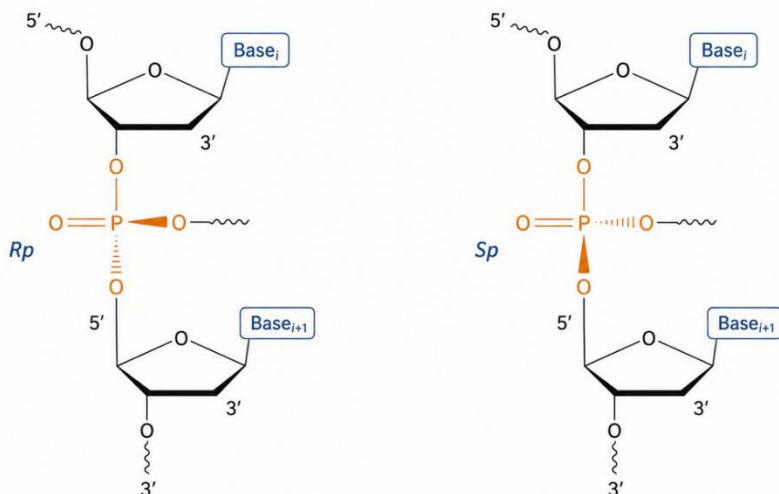


## Chirality



Substitution at the phosphorus center introduces chirality into the backbone, generating Rp and Sp stereoisomers. In phosphorothioate backbones, each linkage can adopt either configuration, resulting in large mixtures of diastereomers in conventionally synthesized oligonucleotides. Chirality can influence multiple aspects of oligonucleotide behavior. Rp linkages are often associated with improved RNase H recognition and RNA binding, whereas Sp linkages confer greater resistance to nuclease degradation. These opposing effects suggest that stereochemical control could, in principle, enable fine-tuning of backbone properties beyond chemistry alone. Recent advances in stereocontrolled synthesis have enabled the generation of stereodefined oligonucleotides, allowing systematic exploration of these effects.

| ASO        | Stereo pattern        | Tm (°C) | Percent knockdown / remaining at day 5 (%) |
|------------|-----------------------|---------|--------------------------------------------|
| Mipomersen | Stereorandom          | 80.2    | 67                                         |
| WW-1       | Rp all                | 84.7    | 34                                         |
| WW-2       | Sp all                | 74.7    | 97                                         |
| WW-3       | Sp-rich mixed pattern | 78.8    | 92                                         |
| WW-4       | Rp-rich mixed pattern | 80.0    | 55                                         |
| WW-5       | Mixed Rp/Sp pattern   | 81.6    | 90                                         |
| WW-6       | Mixed Rp/Sp pattern   | 78.3    | 55                                         |
| WW-7       | Mixed Rp/Sp pattern   | 68.2    |                                            |

Iwamoto, N., Butler, D.C.D., Svrtikapa, N., Mohapatra, S., Zlatev, I., Sah, D.W.Y., Meena, Standley, S.M., Lu, G., Apponi, L.H., Frank-Kamenetsky, M., Zhang, J.J., Vargeese, C. and Verdine, G.L., 2017. *Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides*. Nature Biotechnology.

A team at Wave Life Sciences demonstrated the importance of backbone stereochemistry in a 2017 study where they showed that a lack of control masked significant differences in biological activity, as individual Rp and Sp configurations at phosphorus profoundly influenced stability, protein interactions, and RNase H-mediated cleavage (Iwamoto et al., 2017 ). By developing methods to produce stereopure oligonucleotides with defined chirality patterns, the team demonstrated improved, in vivo, potency, and greater durability compared to conventional stereorandom ASOs (Iwamoto et al., 2017 ). This helped establish a key platform technology for Wave Therapeutics and laid the foundation for compounds such as their DMD compound WVE-N531.

### Stereo-pure ASO WVE-N531



WVE-N531 is a stereopure antisense oligonucleotide (ASO) designed to modulate splicing of the dystrophin pre-mRNA for the treatment of Duchenne muscular dystrophy (DMD). It is a 20-mer single-stranded RNA-like molecule built from a precisely controlled pattern of 2'-fluoro (2'-F) and 2'-O-methyl (2'-OMe) sugar modifications, combined with a chirally defined backbone containing mostly “Sp” phosphorothioate linkages and a smaller number of Rp phosphoramidate linkages, along with a phosphodiester. This stereochemical control is a key feature of Wave Life Sciences’ design approach, as it can influence binding affinity, stability, and protein interactions. WVE-N531 binds to dystrophin pre-mRNA and promotes exon 53 skipping, enabling restoration of a shorter but functional dystrophin protein in patients with certain mutations.

Despite its conceptual appeal, experimental evidence indicates that the benefits of stereochemical control is less obvious than it may seem at first. Systematic studies by a group at Ionis pharmaceuticals demonstrate that while PS chirality modulates RNase H1 cleavage patterns in a predictable manner, these changes do not translate into improved potency or therapeutic index. Stereodefined oligonucleotides exhibit target knockdown comparable to stereorandom mixtures, with no configuration consistently outperforming the parent compounds. Indeed, fully Sp-modified backbones can exhibit substantial losses in activity.

At the mechanistic level, introduction of defined Rp and Sp linkages can bias cleavage site selection and, in some cases, produce dominant cleavage positions. Notably, such effects can be achieved with minimal stereochemical intervention, such as incorporation of a single RpSp step

within the gap region. However, these alterations in enzymatic processing do not result in enhanced cellular or in vivo efficacy, underscoring a disconnect between biochemical precision and pharmacological outcome (Wan WB, Seth PP., 2016).